

Optics

(Scheduled Event - 30 Minutes)



Description: Participants will be assessed on their understanding of the properties and behavior of light, including reflection, refraction, image formation, color, and optical systems, as well as their ability to apply these principles in a hands-on activity involving mirrors and a laser beam.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Teams must bring:

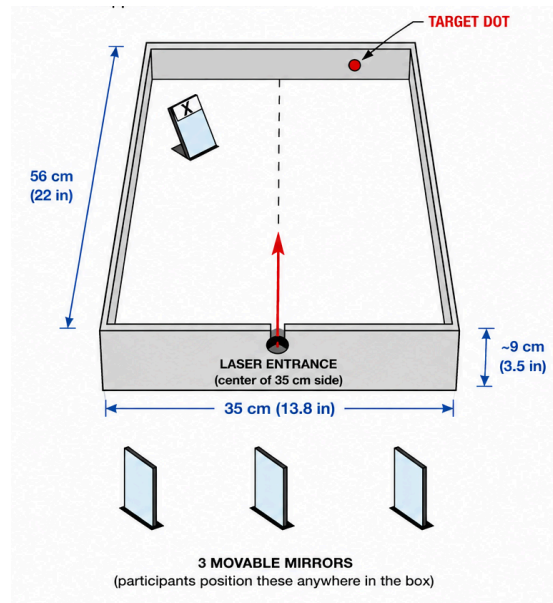
- Pen/Pencil

Teams may bring:

- Four-function calculator
- Protractor & Ruler
- A notecard for use during their laser shoot time to hold in front of the mirrors and aid in seeing where the beam goes when the laser is on.

Laser Shoot (provided at the event):

- A flat-bottom, open box with inner dimensions of approximately 56 cm (22 in) x 35 cm (13.8 in) and a wall height of approximately 9 cm (3.5 in).
- One Class 2 laser pointer mounted through the approximate center of a 35 cm side, aimed into the box along the midline.
- 4 mirrors
 - Each mirror will be mounted vertically, with a width of 5-8 cm (2-3 in).
 - One mirror will be pre-stationed anywhere and at any angle in the box. This pre-stationed mirror will be marked with an "X" and is not to be moved by the participants.
- Somewhere on the wall opposite the entrance of the laser, a target dot will be taped.



Safety Requirements: None

The Competition: This event includes a written test and a hands-on mirror-positioning activity to direct a laser beam toward a target. Students will be taken from the written test to participate in the laser shoot activity and will return back to their test at the conclusion of their laser shoot time.

Content topics will include but are not limited to:

1. Fundamentals of Light

- Describe the properties of light including transmission, absorption, reflection, and refraction.

- Distinguish between transparent, translucent, and opaque materials.
- Explain how light interacts with matter.

2. Reflection and Mirrors

- Apply the law of reflection.
- Describe image formation by plane, concave, and convex mirrors
- Determine image characteristics including location, orientation, and relative size.

3. Refraction and Lenses

- Explain refraction and its causes.
- Describe image formation by convex and concave lenses.
- Apply simple ray diagrams to determine image characteristics.
- Relate refraction to common phenomena such as magnification and apparent bending of light.

4. Color and the Electromagnetic Spectrum

- Describe visible light as part of the electromagnetic spectrum.
- Relate wavelength and frequency to color.
- Distinguish between additive and subtractive color mixing.
- Explain how objects absorb and reflect light to produce color.

5. Optical Systems and Applications

- Describe the operation of common optical devices including magnifying glasses, microscopes, telescopes, cameras, eyeglasses, and fiber optics.
- Explain how optical principles are applied in everyday technologies.

6. Laser Shoot Activity - 2 minutes

- The angle and location of the stationary mirror will not be revealed until the competitors come to the laser shoot. The location of the target will also be revealed at that time.
- Teams must use at least two mirrors, including the stationary mirror, to direct the laser beam to the target. If desired, the team may use additional mirrors. For a mirror to count as being “used”, the laser beam must reflect off of the mirror; it is not enough for the mirror to simply be present in the box.
- Teams will have a total of two minutes to complete the activity. Within this time, and without disturbing the stationary mirror, teams may adjust movable mirrors before the laser is activated. Teams may use their own protractor when setting additional mirrors.
- The Event Supervisor or designee will activate the laser when asked. Each time the team asks the Event Supervisor to turn on the laser counts as one “try”. Mirrors may not be touched or moved when the laser is on.
 - If the laser does not reflect off the stationary mirror and one additional mirror, no information is recorded, no matter how close to the target the beam falls. It is not a valid attempt, but it still counts as a “try”.
 - If the laser beam hits the target on a valid attempt, the Event Supervisor will stop and record the time used and the number of tries the team attempted to hit the target. The team’s laser shoot is complete.
 - If the laser does not hit the target on a valid attempt, the laser will be turned off and the Event Supervisor will mark with a piece of tape along the wall where the beam hit. The team may elect to make another try at hitting the target.
- When the team decides that they have gotten close enough to the target and does not want to make any more “tries” or when 2 minutes have passed (whichever comes first), the Event Supervisor will measure the distance (to the closest tenth of a centimeter) of the team’s closest try to the target. If the closest try is on a wall other than the wall the Target Point is on, the measurement is the sum of the straight-line measurements along the walls of the box (in other words - it is not a point-to-point measurement). The Event Supervisor will record that distance and the total number of “tries” the team made during their time at the laser shoot.

Scoring:

$$\text{Final Score (FS)} = \text{Assessment Score} + \text{Accuracy Score} + \text{“Tries” Score}$$

- Assessment score = $(\text{TAS} \div \text{CAS}) \times 50$; rounded to the 100th place value; maximum 50 points
 - Team Assessment Score (TAS) = team's score on the assessment
 - Competition Assessment Score (CAS) = the highest assessment score for all teams
- Accuracy Score = $(\text{TA} \div \text{CA}) \times 30$
 - Accuracy = $30 - \text{distance measured}$, the lowest Accuracy is 0
 - Competition Accuracy (CA) = the lowest Accuracy over all teams
 - Team Accuracy (TA) = the team's Accuracy
- “Tries” Score = $(\text{CT} \div \text{TT}) \times 20$
 - Competition “Tries” (CT) = the lowest number of “Tries” for all teams
 - Team “Tries” (TT) = the team's number of “Tries”

Scoring Example:

- Competition Assessment Score (CAS) = 15. The team scores 9 on the assessment.
Assessment Score for the team = $(9 \div 15) \times 50 = 30$ points
- Team was 10 cm from the target, thus Team's Accuracy = $30 - 10 = 20$ (TA)
Competition Accuracy (CA) = 25 (the closest team was 5 cm from the target)
Accuracy Score = $(20 \div 25) \times 30 = 24$ points
- The team took 5 tries (TT). The competition “tries” (CT) was 4.
“Tries” Score = $(4 \div 5) \times 20 = 16$ points
- Final Score (FS) = $30 + 24 + 16 = 70$ points

Tiebreakers:

- 1st - the fastest time to hit the target (if the target was hit)
- 2nd - best accuracy score
- 3rd - selected questions from the written assessment

Resources:

- [Optics](#)
- [Light Waves & Colors](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles